Lesson 3.1 - Statistics Formulas and Equations

> **Video:** Statistics - Formulas and Equations by The Organic Chemistry Tutor **Chapter:** 3 - Central Tendency

A reference sheet of the basic formulas used at the start of a statistics course. Later lessons in this chapter work through each one in detail.

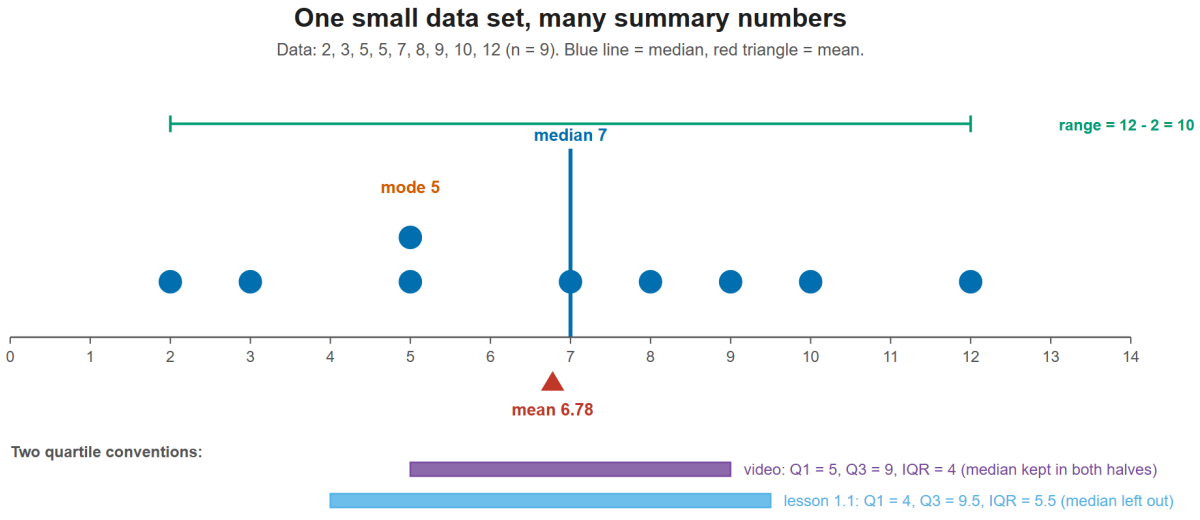
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1. Median, mode, range, quartiles

Example data ($n = 9$, already sorted): 2, 3, 5, 5, 7, 8, 9, 10, 12

Quantity	Definition	Value
Median (Q_2, 50th percentile)	Middle value	7
Mode	Most frequent value	5 (appears twice)
Range	max − min	$12 - 2 = 10$
Q_1 (25th percentile)	Median of the lower half	5 (video)
Q_3 (75th percentile)	Median of the upper half	9 (video)
IQR	$Q_3 - Q_1$	$9 - 5 = 4$

> **Quartile conventions differ.** The video's $Q_1 = 5$ and $Q_3 = 9$ come from **including** the median 7 in both halves (2 3 5 **5** 7 and 7 8 **9** 10 12). The Lesson 1.1 method **excludes** the median, giving halves 2 3 5 5 and 8 9 10 12, so $Q_1 = 4$ and $Q_3 = 9.5$. Use whichever method your course uses, and use it consistently.



The example data as a dot plot with the median (7), mode (5), mean (6.78) and range (10) marked. The two bars at the bottom show why quartiles can differ between textbooks: keeping the median in both halves gives Q1 = 5 and Q3 = 9, while leaving it out gives Q1 = 4 and Q3 = 9.5.

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2. Arithmetic (sample) mean

$$\bar{x} = \frac{\sum x}{n} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

Example: $\bar{x} = \dfrac{2+3+5+5+7+8+9+10+12}{9} = \dfrac{61}{9} \approx 6.78$

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3. Four kinds of mean

For two numbers a and b

Mean	Formula
Arithmetic (AM)	$\dfrac{a+b}{2}$
Geometric (GM)	$\sqrt{ab}$
Harmonic (HM)	$\dfrac{2}{\frac{1}{a}+\frac{1}{b}} = \dfrac{2ab}{a+b}$
Root mean square (RMS)	$\sqrt{\dfrac{a^2+b^2}{2}}$

For any two positive numbers:

$$\mathrm{HM} \le \mathrm{GM} \le \mathrm{AM} \le \mathrm{RMS}, \qquad \mathrm{GM} = \sqrt{\mathrm{AM} \cdot \mathrm{HM}}$$

Quick check with $a = 4$, $b = 9$: AM = 6.5, GM = 6, HM = $72/13 \approx 5.54$, RMS = $\sqrt{48.5} \approx 6.96$. The order is 5.54 ≤ 6 ≤ 6.5 ≤ 6.96 ✓, and $\sqrt{6.5 \times 5.54} = 6$ ✓.

For n numbers

Mean	Formula
Arithmetic	$\bar{x} = \dfrac{x_1 + x_2 + \cdots + x_n}{n}$
Geometric	$(x_1 \cdot x_2 \cdots x_n)^{1/n}$
Harmonic	$\dfrac{n}{\frac{1}{x_1}+\frac{1}{x_2}+\cdots+\frac{1}{x_n}}$
Root mean square	$\sqrt{\dfrac{x_1^2 + x_2^2 + \cdots + x_n^2}{n}}$
Weighted	$\bar{x}_w = \dfrac{\sum wx}{\sum w} = \dfrac{w_1x_1 + w_2x_2 + \cdots + w_nx_n}{w_1 + w_2 + \cdots + w_n}$

With $n = 2$ the geometric-mean formula becomes $(ab)^{1/2} = \sqrt{ab}$.

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4. Sample vs population symbols

	Sample	Population
Mean	$\bar{x}$	μ
Size	n	N
Standard deviation	s	σ
Variance	s^2	σ^2

Example: a city has $N = 10{,}000$ residents, but a study surveys $n = 300$ of them. n and N are different numbers.

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5. Standard deviation and variance

	Sample	Population
Standard deviation	$s = \sqrt{\frac{\sum(x-\bar{x})^2}{n-1}}$	$\sigma = \sqrt{\frac{\sum(x-\mu)^2}{N}}$
Variance	$s^2 = \frac{\sum(x-\bar{x})^2}{n-1}$	$\sigma^2 = \frac{\sum(x-\mu)^2}{N}$

- The **sample** formulas divide by $n-1$. The **population** formulas divide by N.
- Variance = (standard deviation)^2, the same formula without the square root.

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6. Coefficient of variation (CV)

$$\mathrm{CV}_{\text{sample}} = \frac{s}{\bar{x}} \times 100\%, \qquad \mathrm{CV}_{\text{population}} = \frac{\sigma}{\mu} \times 100\%$$

The CV expresses the standard deviation as a **percentage of the mean**. A CV of 0.25 (25%) means the standard deviation is 25% of the mean.

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7. Mean absolute deviation (MAD)

$$\mathrm{MAD} = \frac{\sum |x - \bar{x}|}{n}$$

It looks like the standard deviation, but it uses **absolute values instead of squares** and has **no square root**.

Key takeaways

- Know the four means (AM, GM, HM, RMS), their order HM ≤ GM ≤ AM ≤ RMS, and the weighted mean.
- Sample formulas use $\bar{x}$, n, s and divide variance by $n-1$. Population formulas use μ, N, σ and divide by N.
- CV = SD ÷ mean × 100%. MAD = average absolute distance from the mean.

Lesson 3.2 - Mean, Median, Mode, and Range

> **Video:** Mean, Median, Mode, and Range - How To Find It! by The Organic Chemistry Tutor **Chapter:** 3 - Central Tendency

Definitions

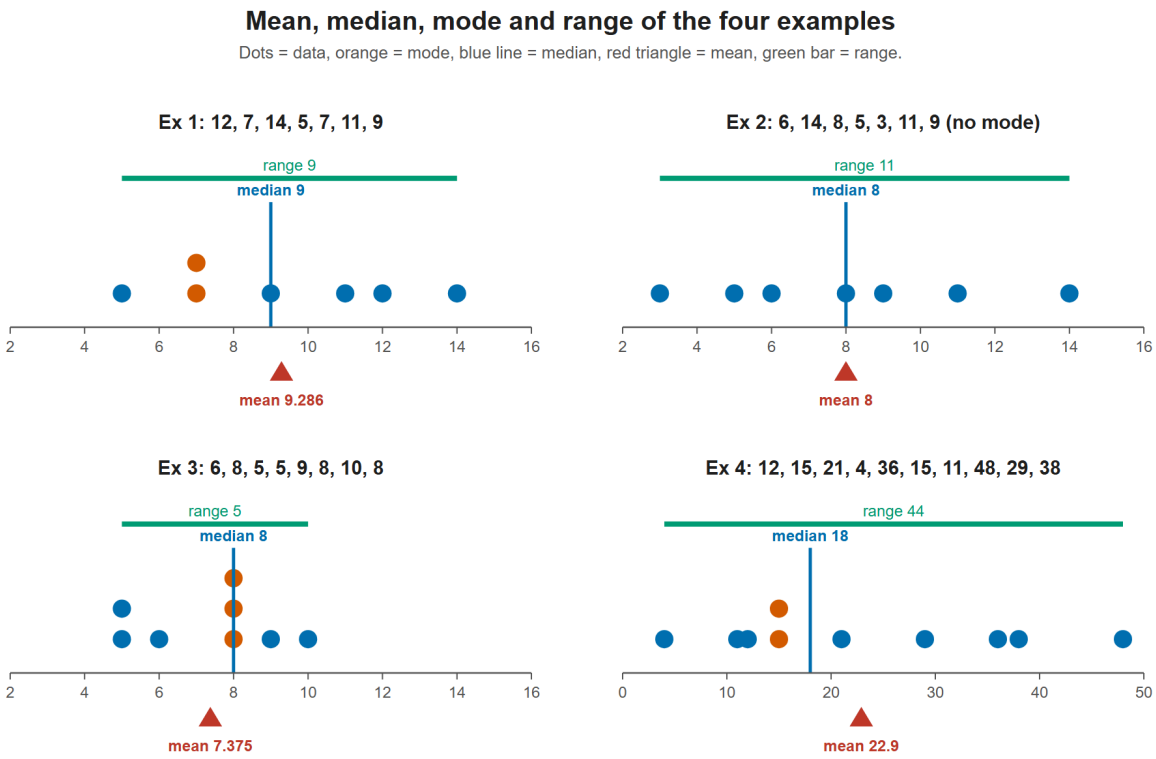
Measure	Rule
Mean (arithmetic mean, average)	Sum of the values ÷ number of values
Median	Middle value of the **sorted** data. Even n: average of the two middle values
Mode	Value that appears most often. It can have **no mode** or **several modes**
Range	Highest − lowest

Finding the median by hand: sort the data, then cross off the first and last values, then the next pair, and so on, until one value (odd n) or two values (even n) are left.

Example 1 - odd n (7 values)

Data: 12, 7, 14, 5, 7, 11, 9 → sorted: 5, 7, 7, **9**, 11, 12, 14

- Mean: $\frac{65}{7} \approx 9.29$
- Median: **9**
- Mode: **7** (the only value that appears twice)
- Range: $14 - 5 = 9$



All four examples on number lines. In Example 2 every value appears once, so there is no mode. In Example 4 the large values (36, 38, 48) pull the mean (22.9) well to the right of the median (18).

Example 2 - odd n, no mode

Data: 6, 14, 8, 5, 3, 11, 9 → sorted: 3, 5, 6, **8**, 9, 11, 14

- Mean: $\frac{56}{7} = 8$
- Median: **8**
- Mode: **none**, since every value appears once
- Range: $14 - 3 = 11$

Example 3 - even n (8 values)

Data: 6, 8, 5, 5, 9, 8, 10, 8 → sorted: 5, 5, 6, **8, 8**, 8, 9, 10

- Mean: $\frac{59}{8} = 7.375$
- Mode: **8** (appears three times)
- Median: the two middle values are both 8 → $\frac{8+8}{2} = 8$
- Range: $10 - 5 = 5$

Example 4 - even n (10 values)

Data: 12, 15, 21, 4, 36, 15, 11, 48, 29, 38 Sorted: 4, 11, 12, 15, **15, 21**, 29, 36, 38, 48

- Mean: $\frac{229}{10} = 22.9$
- Mode: **15**
- Median: $\frac{15 + 21}{2} = 18$, the midpoint of the two middle values
- Range: $48 - 4 = 44$

Key takeaways

- **Sort first.** It makes the median, mode and range easy.
- Odd n: the median is a data value. Even n: the median is the **average of the two middle values**, which may not be in the data.
- A data set can have **no mode** (all values different) or **more than one** mode.

Lesson 3.3 - How to Calculate the Median Using a Formula

> **Video:** How To Calculate The Median Using a Formula by The Organic Chemistry Tutor **Chapter:** 3 - Central Tendency

The formulas

First **sort the data** in ascending order. Let a_k be the k-th value in the sorted list.

When n is odd:

$$\text{median} = a_{\frac{n+1}{2}}$$

When n is even:

$$\text{median} = \frac{a_{\frac{n}{2}} + a_{\frac{n}{2}+1}}{2}$$

This is the average of the two middle terms.

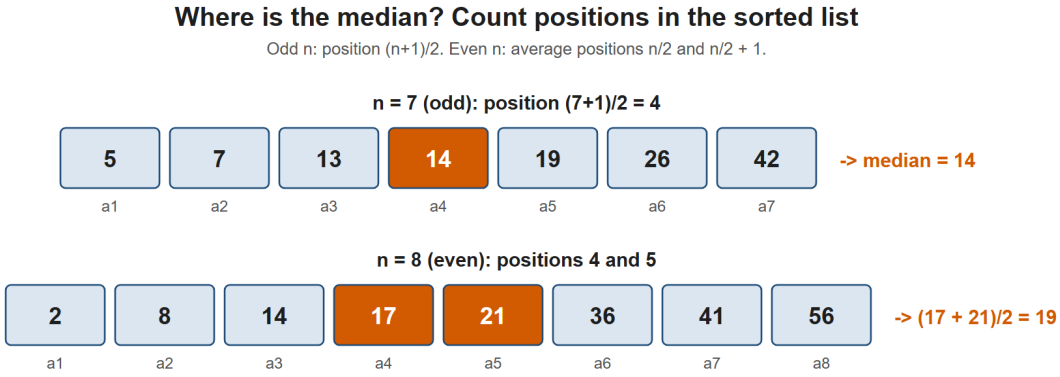
The formula tells you **which position** holds the median, which is handy when n is large.

Example 1 - odd n

Data: 7, 14, 5, 19, 26, 42, 13

Sorted: 5, 7, 13, **14**, 19, 26, 42 ($n = 7$)

$$\frac{n+1}{2} = \frac{8}{2} = 4 \;\Rightarrow\; \text{median} = a_4 = 14$$



The formula only tells you a position in the sorted list. With 7 values the 4th box is the median. With 8 values there is no single middle box, so the 4th and 5th are averaged.

Example 2 - even n

Data: 8, 21, 14, 36, 17, 2, 56, 41

Sorted: 2, 8, 14, **17, 21**, 36, 41, 56 ($n = 8$)

$$\frac{n}{2} = 4, \quad \frac{n}{2} + 1 = 5 \;\Rightarrow\; \text{median} = \frac{a_4 + a_5}{2} = \frac{17 + 21}{2} = 19$$

Key takeaways

- Always sort first. The formula gives a **position**, not a value.
- Odd n: one middle term at position $(n + 1)/2$.
- Even n: average the terms at positions $n/2$ and $n/2 + 1$.

Lesson 3.4 - How to Calculate the Range, Midrange, and Coefficient of Range

> **Video:** How To Calculate the Range, MidRange, and Coefficient of Range by The Organic Chemistry Tutor
> **Chapter:** 3 - Central Tendency

Formulas

Let H = highest value and L = lowest value.

Quantity	Formula	Meaning
Range	$H - L$	Total spread of the data
Midrange	$\frac{H + L}{2}$	Halfway point between the extremes
Coefficient of range (CR)	$\frac{H - L}{H + L}$	Range relative to the size of the values (no units)

Example

In the given data set, the lowest value is $L = 5$ and the highest is $H = 45$.

(a) Range

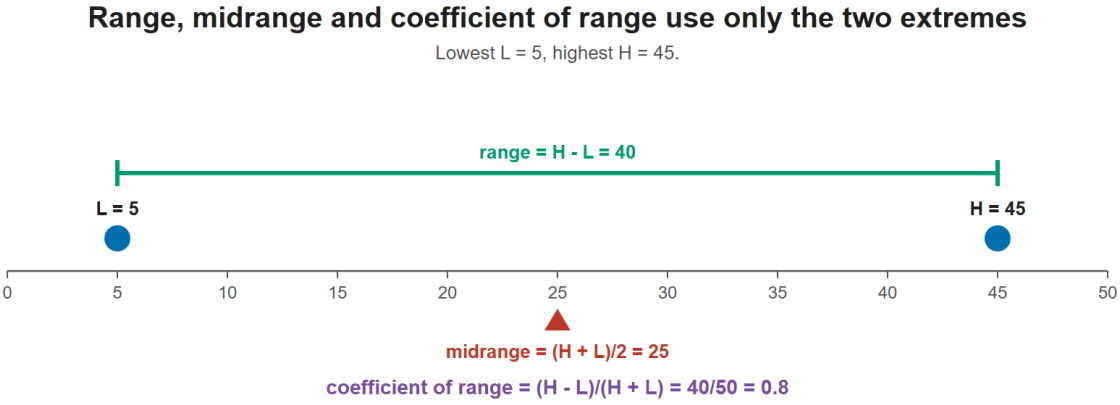
$$45 - 5 = 40$$

(b) Midrange

$$\frac{45 + 5}{2} = \frac{50}{2} = 25$$

(c) Coefficient of range

$$\text{CR} = \frac{45 - 5}{45 + 5} = \frac{40}{50} = \frac{4}{5} = 0.8$$



All three measures come from just the two extreme values: the range is the distance between them, the midrange is the point halfway between them, and the coefficient of range compares the range with their sum.

Key takeaways

- All three use only the **highest** and **lowest** values, so they are sensitive to outliers.
- The midrange is a measure of center. The range and coefficient of range are measures of spread.
- The coefficient of range has no units, so it can compare the spread of data sets measured on different scales.

Lesson 3.5 - How to Find the Missing Value Given the Mean

> **Video:** How To Find The Missing Value Given The Mean by The Organic Chemistry Tutor **Chapter:** 3 - Central Tendency

Idea

If you know the **target mean** and all values except one, you can solve for the missing value.

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n} \quad \Longrightarrow \quad x_n = n\,\bar{x} - (x_1 + x_2 + \cdots + x_{n-1})$$

In words: **missing value = (number of values × desired mean) − (sum of the known values).**

$n\bar{x}$ is the **total** the scores must add up to. Subtract what you already have, and the remainder is the value you need.

Example 1 - what score is needed on exam 5?

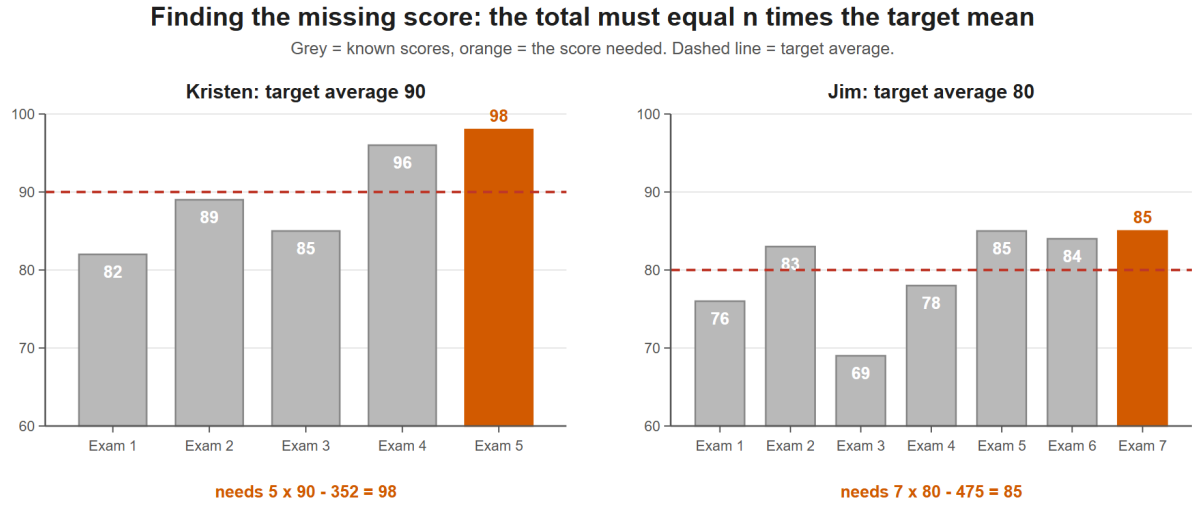
Kristen scored **82, 89, 85, 96** on her first four chemistry exams. What does she need on the fifth exam to average **90**?

$$90 = \frac{82 + 89 + 85 + 96 + x_5}{5}$$

Multiply both sides by 5:

$$450 = 352 + x_5 \quad \Rightarrow \quad x_5 = 450 - 352 = 98$$

She needs a **98**.



The needed score (orange) has to make up for the scores below the target line and use up the room left by the scores above it, so that the bars balance at the target average.

Example 2 - what score is needed on exam 7?

Jim's first six exam scores are **76, 83, 69, 78, 85, 84**. What does he need on the seventh exam to average **80**?

$$x_7 = 7(80) - (76 + 83 + 69 + 78 + 85 + 84) = 560 - 475 = 85$$

He needs an **85**.

Key takeaways

- Required total = $n\times$ target mean.
- Missing value = required total − sum of the known values.
- If the answer is above the maximum possible score (e.g. over 100), the target average cannot be reached.

Lesson 3.6 - How to Find the Weighted Mean

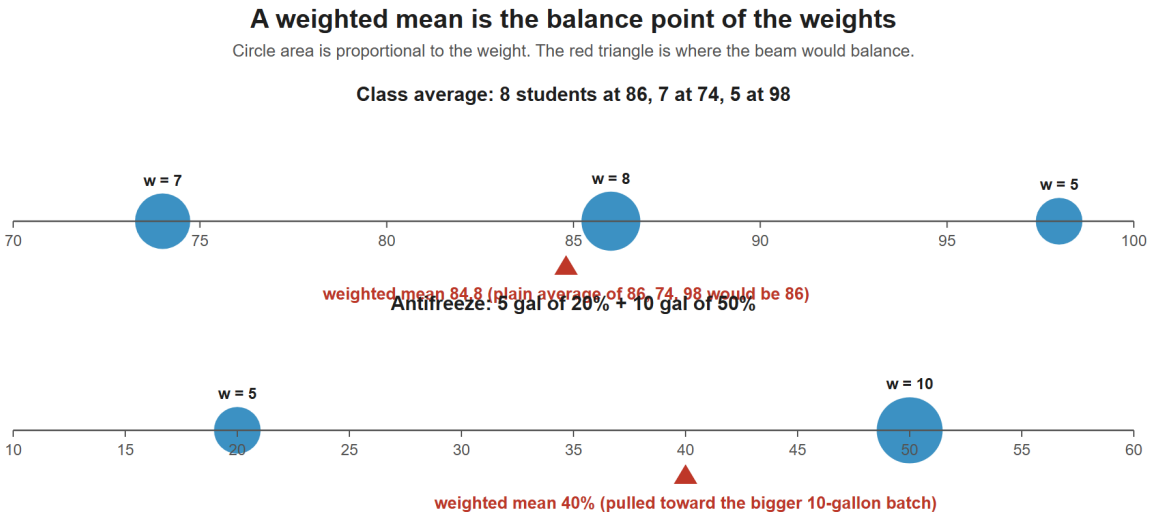
> **Video:** How To Find The Weighted Mean and Weighted Average In Statistics by The Organic Chemistry Tutor **Chapter:** 3 - Central Tendency

Formula

$$\bar{x}_w = \frac{\sum wx}{\sum w} = \frac{w_1x_1 + w_2x_2 + \cdots + w_nx_n}{w_1 + w_2 + \cdots + w_n}$$

- x = the values you want to average.
- w = the **weight** of each value: a frequency, a count of people, a percentage, credit hours, a volume, ...
- If the weights are percentages that add to 100% (or decimals that add to 1), the denominator is 1 and you only need the numerator.

> **How to decide which is x and which is w:** x is the thing you want the *average of*. w is *how much each one counts*.



Picture each value as a weight on a beam. The weighted mean is where the beam balances, so it moves toward the heavier weights: 84.8 instead of 86 for the class averages, and 40% instead of 35% for the antifreeze mixture.

Example 1 - a weighted mean with frequencies as weights

Data (10 values): one 10, one 12, four 16s, three 20s, one 24.

Ordinary mean: $\frac{170}{10} = 17$

Weighted mean, using frequencies as weights:

$$\bar{x}_w = \frac{1(10) + 1(12) + 4(16) + 3(20) + 1(24)}{1 + 1 + 4 + 3 + 1} = \frac{170}{10} = 17$$

The two give the same answer: a mean computed from a frequency table **is** a weighted mean.

Example 2 - combining group averages

In a class of 20: 8 students averaged 86, 7 averaged 74, and 5 averaged 98. What is the class average?

$$\bar{x}_w = \frac{8(86) + 7(74) + 5(98)}{8 + 7 + 5} = \frac{688 + 518 + 490}{20} = \frac{1696}{20} = 84.8$$

> Averaging 86, 74 and 98 directly gives 86, which is wrong because the groups are different sizes.

Example 3 - percentages as weights

At a college, 20% of students average 140 lb, 35% average 160 lb, 30% average 175 lb, and 15% average 195 lb.

$$\bar{x}_w = 0.20(140) + 0.35(160) + 0.30(175) + 0.15(195) = 28 + 56 + 52.5 + 29.25 = 165.75 \text{ lb}$$

(The weights add to 1, so there is nothing to divide by.)

Example 4 - final semester grade

Category	Weight	John	Kelly
Homework	15%	92	100
Quizzes	10%	74	82
Lab	20%	83	95
Tests	25%	76	70
Final exam	30%	88	76

John: $0.15(92) + 0.10(74) + 0.20(83) + 0.25(76) + 0.30(88) = 13.8 + 7.4 + 16.6 + 19 + 26.4 = \mathbf{83.2}$

Kelly: $0.15(100) + 0.10(82) + 0.20(95) + 0.25(70) + 0.30(76) = 15 + 8.2 + 19 + 17.5 + 22.8 = \mathbf{82.5}$

Kelly did better on homework, quizzes and lab, but those are only **45%** of the grade. John did better on tests and the final, which are **55%**, so his weighted grade is slightly higher.

Example 5 - grade point average (GPA)

Points: A = 4, B = 3, C = 2. Weights = credit hours.

Course	Credit hours w	Grade	Points x	$w \cdot x$
Chemistry	3	B	3	9
Physics	3	C	2	6
Lab	1	A	4	4
Calculus	4	A	4	16
English	3	B	3	9
Total	**14**			**44**

$$\text{GPA} = \frac{44}{14} \approx 3.14$$

Example 6 - mixing solutions

Rachel mixes **5 gal of 20%** antifreeze with **10 gal of 50%** antifreeze. Is the mixture closer to 20% or 50%?

- An equal-weight average would be 35%.
- The 50% solution has more volume, so it carries more weight. The answer must be between 35% and 50%.

$$C = \frac{5(20\%) + 10(50\%)}{5 + 10} = \frac{100 + 500}{15} = \frac{600}{15} = 40\%$$

The mixture is **40%**, closer to 50% as predicted. The gallons cancel, leaving a percentage.

Key takeaways

- Weighted mean = $\sum wx / \sum w$.
- Use it whenever the values **don't count equally**: group averages, course grades, GPA, mixtures.
- The weighted mean is pulled toward the values with the **largest weights**.

Lesson 3.7 - The Mean Formulas

> **Video:** Arithmetic Mean, Geometric Mean, Weighted Mean, Harmonic Mean, Root Mean Square Formula by The Organic Chemistry Tutor **Chapter:** 3 - Central Tendency

This lesson covers five kinds of "average", how each relates to a type of sequence, and when to use each.

Mean	Formula (n values)	Two values a, b
Arithmetic	$\frac{x_1 + \cdots + x_n}{n}$	$\frac{a+b}{2}$
Geometric	$(x_1 x_2 \cdots x_n)^{1/n}$	$\sqrt{ab}$
Weighted	$\frac{\sum wx}{\sum w}$	—
Harmonic	$\frac{n}{\frac{1}{x_1} + \cdots + \frac{1}{x_n}}$	$\frac{2}{\frac{1}{a}+\frac{1}{b}} = \frac{2ab}{a+b}$
Root mean square	$\sqrt{\frac{x_1^2 + \cdots + x_n^2}{n}}$	$\sqrt{\frac{a^2+b^2}{2}}$

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1. Arithmetic mean and arithmetic sequences

An **arithmetic sequence** adds the same **common difference** d each time.

Example: 2, 5, 8, 11, 14, 17, 20 ($d = 3$)

- Mean of all seven terms: $\frac{77}{7} = 11$, which is the **middle term** a_4.
- The arithmetic mean of two terms equidistant from a middle term gives that middle term:
  - $\frac{a_1 + a_3}{2} = \frac{2+8}{2} = 5 = a_2$
  - $\frac{a_1 + a_5}{2} = \frac{2+14}{2} = 8 = a_3$
  - $\frac{a_3 + a_7}{2} = \frac{8+20}{2} = 14 = a_5$

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2. Geometric mean and geometric sequences

A **geometric sequence** multiplies by the same **common ratio** r each time.

Example: 3, 6, 12, 24, 48, 96, 192 ($r = 2$)

- $\sqrt{a_1 a_3} = \sqrt{3 \cdot 12} = \sqrt{36} = 6 = a_2$
- $\sqrt{a_1 a_5} = \sqrt{3 \cdot 48} = \sqrt{144} = 12 = a_3$
- Geometric mean of all seven terms:

  $$(3 \cdot 6 \cdot 12 \cdot 24 \cdot 48 \cdot 96 \cdot 192)^{1/7} = (4{,}586{,}471{,}424)^{1/7} = 24 = a_4$$

  This is the middle term again.

With $n = 2$, $(x_1 x_2)^{1/2}$ is just $\sqrt{ab}$.

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3. Weighted mean

$$\bar{x}_w = \frac{w_1 x_1 + w_2 x_2 + \cdots + w_n x_n}{w_1 + w_2 + \cdots + w_n}$$

Use it when the data points are **not equally important**: center of mass, average atomic mass, concentration of a mixture.

Example - mixing salt solutions. 150 mL of 20% NaCl is mixed with 50 mL of 40% NaCl. Here x is the concentration and w is the volume:

$$C = \frac{C_1 V_1 + C_2 V_2}{V_1 + V_2} = \frac{20(150) + 40(50)}{150 + 50} = \frac{3000 + 2000}{200} = 25\%$$

An equal-weight average would give 30%. There is more of the 20% solution, so the result is pulled toward 20%. With equal volumes (100 mL each), the answer would be exactly 30%.

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4. Harmonic mean and harmonic sequences

The harmonic mean is the **reciprocal of the average of the reciprocals**:

$$H = \left(\frac{\frac{1}{x_1} + \cdots + \frac{1}{x_n}}{n} \right)^{-1} = \frac{n}{\sum \frac{1}{x}}$$

From the definition to the product form

Put the reciprocals over a common denominator.

Two values a, b (common denominator ab):

$$\frac{1}{a} + \frac{1}{b} = \frac{b + a}{ab} \quad \Rightarrow \quad H = \frac{2}{\frac{a+b}{ab}} = \frac{2ab}{a + b}$$

Three values a, b, c (common denominator abc):

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{bc + ac + ab}{abc} \quad \Rightarrow \quad H = \frac{3abc}{ab + bc + ca}$$

General n**:** each $\frac{1}{x_j}$ becomes the product of all the *other* values over the full product, so

$$H = \frac{n \prod_{i=1}^{n} x_i}{\sum_{j=1}^{n} \prod_{i \neq j} x_i}$$

For $n = 2$, "the product of the others" is just one number (b or a), so the denominator collapses to $a + b$.

> **Warning:** $\frac{n \prod x_i}{\sum x_i}$ is **only** correct for $n = 2$. Check with 1, 2, 4:
>
> - True: $H = \frac{3}{1 + 0.5 + 0.25} = \frac{3}{1.75} \approx 1.714$
> - Correct product form: $\frac{3(8)}{2 + 4 + 8} = \frac{24}{14} \approx 1.714$
> - Wrong shortcut: $\frac{3(8)}{1 + 2 + 4} = \frac{24}{7} \approx 3.43$, larger than the arithmetic mean (2.33), which a harmonic mean can never be.

> For $n > 2$, just use $H = \dfrac{n}{\sum \frac{1}{x}}$.

A **harmonic sequence** consists of the reciprocals of an arithmetic sequence:

$$\tfrac{1}{2}, \tfrac{1}{5}, \tfrac{1}{8}, \tfrac{1}{11}, \tfrac{1}{14}, \tfrac{1}{17}, \tfrac{1}{20}$$

(The denominators 2, 5, 8, ... form an arithmetic sequence.)

- $\dfrac{2}{\frac{1}{h_1} + \frac{1}{h_3}} = \dfrac{2}{2+8} = \dfrac{1}{5} = h_2$
- $\dfrac{2}{\frac{1}{h_2} + \frac{1}{h_6}} = \dfrac{2}{5+17} = \dfrac{1}{11} = h_4$
- All seven terms: $\dfrac{7}{2+5+8+11+14+17+20} = \dfrac{7}{77} = \dfrac{1}{11} = h_4$

Application - average speed for equal distances

A car drives **400 mi** from A to B at **80 mph**, then returns **400 mi** at **50 mph**. What is its average speed?

Physics method (total distance ÷ total time):

- Time out: $\dfrac{400}{80} = 5$ h. Time back: $\dfrac{400}{50} = 8$ h.
- $\bar{v} = \dfrac{800}{13} \approx 61.54$ mph

Weighted-mean method (weights = time spent at each speed):

$$\bar{v} = \frac{80(5) + 50(8)}{5+8} = \frac{800}{13} \approx 61.54 \text{ mph}$$

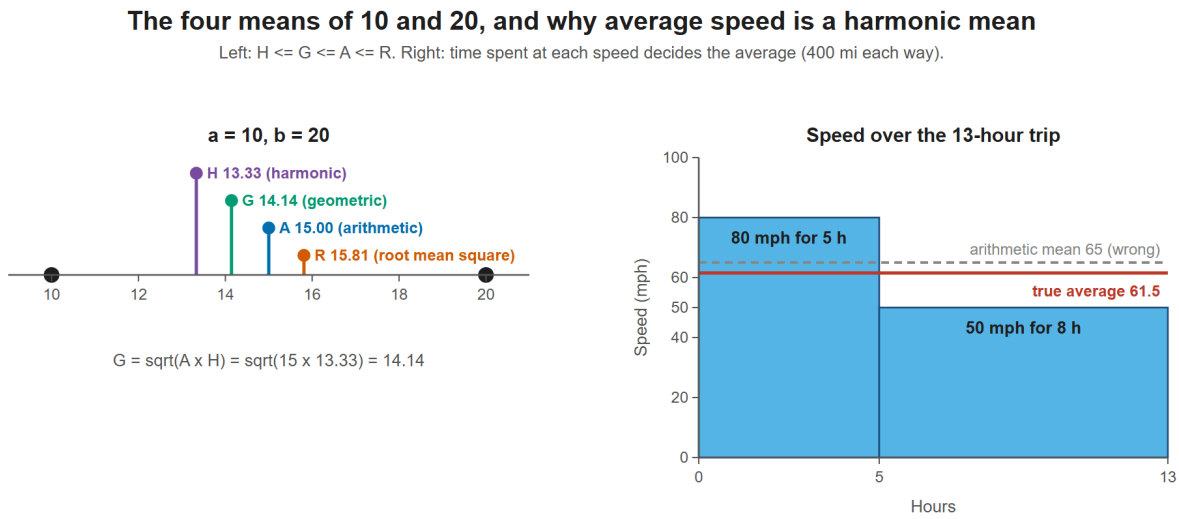
Or using time fractions: $80\left(\frac{5}{13}\right) + 50\left(\frac{8}{13}\right) = 80(0.3846) + 50(0.6154) \approx 61.54$ mph. The car spends about 38.5% of the time at 80 mph and about 61.5% at 50 mph.

Harmonic-mean method:

$$\bar{v} = \frac{2}{\frac{1}{80} + \frac{1}{50}} = \frac{2(80)(50)}{80+50} = \frac{8000}{130} = \frac{800}{13} \approx 61.54 \text{ mph}$$

The arithmetic mean (65 mph) is **wrong** here. The car spends **more time** at the slower speed, so the true average is closer to 50.

> When **equal distances** are covered at different speeds, the average speed is the **harmonic mean** of the speeds.



Left: for 10 and 20 the four means always come in the order H, G, A, R. Right: on the round trip the car spends 8 hours at 50 mph but only 5 hours at 80 mph, so the true average speed (61.5 mph, a harmonic mean) is below the plain average of 65.

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5. Root mean square (RMS)

"Root" = square root, "square" = square the values:

$$x_{\text{RMS}} = \sqrt{\frac{x_1^2 + x_2^2 + \cdots + x_n^2}{n}}$$

Example. 5, 8, 12, 16, 21

$$\sqrt{\frac{25 + 64 + 144 + 256 + 441}{5}} = \sqrt{\frac{930}{5}} = \sqrt{186} \approx 13.64$$

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6. Comparing the four means

Example. $a = 10$, $b = 20$

Mean	Calculation	Value
Harmonic H	$\frac{2(10)(20)}{30} = \frac{40}{3}$	13.33
Geometric G	$\sqrt{200} = 10\sqrt{2}$	14.14
Arithmetic A	$\frac{30}{2}$	15
RMS R	$\sqrt{\frac{100 + 400}{2}} = \sqrt{250} = 5\sqrt{10}$	15.81

$$H \le G \le A \le R$$

Relationships between the three classical means:

$$G = \sqrt{A \cdot H}, \qquad H = \frac{G^2}{A}, \qquad A = \frac{G^2}{H}$$

Check: $\sqrt{15 \cdot \frac{40}{3}} = \sqrt{200} = 10\sqrt{2} = G$ ✓

If you know any two of A, G, H, you can find the third.

Key takeaways

- **Arithmetic mean:** equal weights. It gives the middle term of an arithmetic sequence.
- **Geometric mean:** multiply, then take the n-th root. It gives the middle term of a geometric sequence.
- **Weighted mean:** use when the values carry different importance (volume, time, credits).
- **Harmonic mean:** average of rates over **equal distances** (or equal amounts). It gives the middle term of a harmonic sequence.
- **RMS:** square, average, then square root. It is the largest of the four.
- $H \leq G \leq A \leq R$ and $G^2 = A \cdot H$.

Lesson 3.8 - Formulas of Statistics in Excel

> **Video:** Statistics - Excel by The Organic Chemistry Tutor **Chapter:** 3 - Central Tendency

The video applies Excel's built-in functions to a column of **19 test scores** in cells `A2:A20`, ranging from 52 to 99. Every function has the same pattern: type `=`, the function name, then select the data range.

Basic summary functions

What you want	Excel formula	Result in the video
Sum	`=SUM(A2:A20)`	—
Number of values n	`=COUNT(A2:A20)`	19
Mean	`=AVERAGE(A2:A20)`, or `=SUM(...)/COUNT(...)`	—
Median	`=MEDIAN(A2:A20)`	84
Mode	`=MODE(A2:A20)` (or `=MODE.SNGL`)	86 (the only score that appears twice)
Minimum	`=MIN(A2:A20)`	52
Maximum	`=MAX(A2:A20)`	99
Range	`=MAX(A2:A20)-MIN(A2:A20)`	47

`MIN` and `MAX` are especially useful with large data sets (thousands of rows) where you can't spot the extremes by eye.

Spread: standard deviation and variance

What you want	Sample (usual choice)	Population
Standard deviation	`=STDEV.S(A2:A20)` → 12.46	`=STDEV.P(A2:A20)`
Variance	`=VAR.S(A2:A20)`	`=VAR.P(A2:A20)`

Variance = SD^2, so if the SD is in cell `E10`, then `=E10^2` gives the same value as `VAR.S`.

Quartiles

`=QUARTILE(A2:A20, k)`, where `k` is 0-4. Newer versions also have `QUARTILE.INC` (same as `QUARTILE`) and `QUARTILE.EXC`, which give slightly different answers.

k	Meaning	Equivalent	Video result
0	0th percentile	minimum	52
1	Q_1, 25th percentile		75
2	Q_2, 50th percentile	median	84
3	Q_3, 75th percentile		88.5
4	100th percentile	maximum	99

Percentiles

`=PERCENTILE(A2:A20, p)` (or `PERCENTILE.INC`). Enter p as a **decimal**: the 25th percentile is `0.25`, **not** `25`.

Percentile	Formula	Same as
0th	`=PERCENTILE(A2:A20, 0)`	minimum, `QUARTILE(...,0)`
25th	`=PERCENTILE(A2:A20, 0.25)`	Q_1
40th	`=PERCENTILE(A2:A20, 0.40)`	
50th	`=PERCENTILE(A2:A20, 0.5)`	median, Q_2
75th	`=PERCENTILE(A2:A20, 0.75)`	Q_3
90th	`=PERCENTILE(A2:A20, 0.9)`	(94.4 in the video)
100th	`=PERCENTILE(A2:A20, 1)`	maximum

Deciles

Deciles split the data into **10** equal parts ("deci" = one tenth), just as quartiles split it into 4. Excel has no decile function, so use `PERCENTILE`:

Decile	Formula
D_0	`=PERCENTILE(A2:A20, 0)` (minimum)
D_1	`=PERCENTILE(A2:A20, 0.1)`
D_5	`=PERCENTILE(A2:A20, 0.5)` (median)
D_7	`=PERCENTILE(A2:A20, 0.7)`
D_{10}	`=PERCENTILE(A2:A20, 1)` (maximum)

In general, decile k is `PERCENTILE(range, k/10)`.

Key takeaways

- `SUM`, `COUNT`, `AVERAGE`, `MEDIAN`, `MODE`, `MIN`, `MAX` cover the basics.
- Use `STDEV.S` / `VAR.S` for **samples** and `STDEV.P` / `VAR.P` for **populations**.

- `QUARTILE(range, 0…4)`, `PERCENTILE(range, decimal)`. Deciles = `PERCENTILE(range, k/10)`.
- Excel's quartile and percentile rules can differ slightly from hand methods, because there are several conventions.

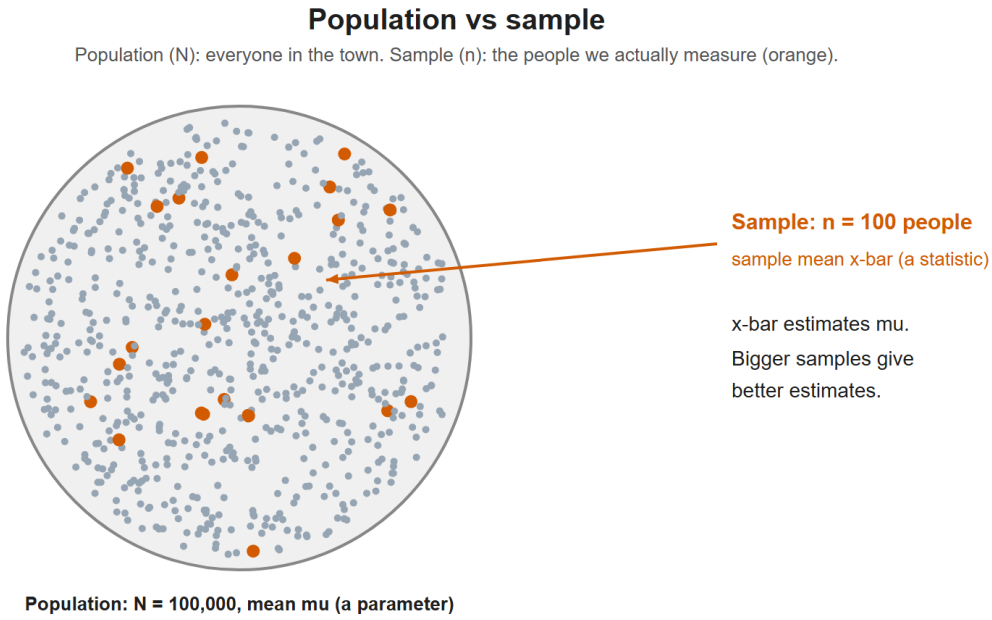
Lesson 3.9 - Sample Mean and Population Mean

> **Video:** Sample Mean and Population Mean by The Organic Chemistry Tutor **Chapter:** 3 - Central Tendency

Population vs sample

- **Population** (size N): everyone of interest. Example: all **100,000** people in city XYZ.
- **Sample** (size n): the part we actually measure. Example: **100** people chosen from the city.

Weighing all 100,000 residents is impractical, so we weigh a sample of 100 and use its average to **estimate** the average weight of the whole city.



The population is everyone (grey dots); the sample is the small group actually measured (orange). The sample mean x-bar is used to estimate the population mean mu.

Formulas

Sample mean (a **statistic**):

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

Population mean (a **parameter**):

$$\mu = \frac{\sum_{i=1}^{N} x_i}{N} = \frac{x_1 + x_2 + \cdots + x_N}{N}$$

The formulas look the same. The only difference is **lowercase** n (sample size) versus **capital** N (population size).

	Symbol	Size	Called a
Sample mean	$\bar{x}$	n	**statistic**
Population mean	μ	N	**parameter**

Bigger samples give better estimates

We want $\bar{x}$ to be as close as possible to μ. **Increasing the sample size** makes $\bar{x}$ a better estimate of μ. Going from $n = 100$ to $n = 1{,}000$ gives a more accurate estimate, but it is ten times as much work.

> **Trade-off:** more accuracy requires more data, and more data requires more effort.

Key takeaways

- $\bar{x}$ (sample, n) is a **statistic**. μ (population, N) is a **parameter**.
- Both are "sum ÷ count". Only the group being averaged differs.
- Larger samples make $\bar{x}$ a more reliable estimate of μ.

Lesson 3.10 - Averages and Uncertainty Calculations

> **Video:** Averages and Uncertainty Calculations by The Organic Chemistry Tutor **Chapter:** 3 - Central Tendency

Idea

Report a set of measurements as **mean ± uncertainty**, where the uncertainty here is **half the range**:

$$u = \frac{\max - \min}{2}, \qquad \text{result} = \bar{x} \pm u$$

On a number line, the mean sits in the middle. Going up by u reaches roughly the top of the data, and going down by u reaches roughly the bottom, so $\bar{x} \pm u$ spans about the whole range.

The uncertainty is usually **rounded to one significant figure**.

> This "half the range" rule is a quick estimate common in lab courses. Later chapters use the standard deviation and the standard error as more formal measures of uncertainty.

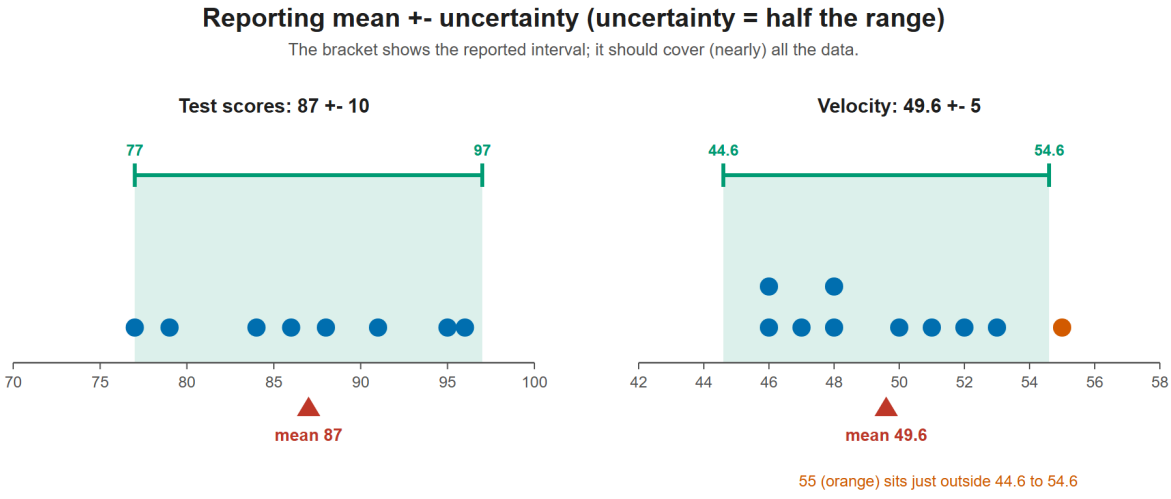
Example 1 - test scores

Scores: 79, 84, 91, 86, 95, 77, 88, 96

- Mean: $\bar{x} = \frac{696}{8} = 87$
- Uncertainty: $u = \frac{96 - 77}{2} = \frac{19}{2} = 9.5 \approx 10$ (1 significant figure)

$$\text{score} = 87 \pm 10$$

Check: $87 - 10 = 77$ and $87 + 10 = 97$, so every score lies inside this interval.



The interval mean +- (half the range) covers nearly all of the data. For the velocities, rounding the uncertainty to 5 leaves the largest value, 55, just outside the interval.

Example 2 - velocity measured once per second

Velocities: 46, 48, 47, 48, 51, 53, 55, 52, 50, 46

- Mean: $\bar{x} = \dfrac{496}{10} = 49.6$
- Uncertainty: $u = \dfrac{55 - 46}{2} = 4.5 \approx 5$

$$v = 49.6 \pm 5$$

This gives the interval 44.6 to 54.6. The maximum 55 sits just outside it, so the summary is close but not perfect.

> **Significant-figure note:** many lab manuals also round the mean to the same decimal place as the uncertainty, giving 50 ± 5.

Key takeaways

- Mean = sum ÷ count. Uncertainty (quick method) = range ÷ 2.
- Round the uncertainty to **one significant figure**.
- Write the result as $\bar{x} \pm u$. The interval should cover nearly all the data.

Lesson 3.11 - How to Calculate the Standard Deviation

> **Video:** How To Calculate The Standard Deviation by The Organic Chemistry Tutor **Chapter:** 3 - Central Tendency

Formula (sample standard deviation)

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

Steps

1. Find the **mean** $\bar{x}$.
2. For each value, find the **deviation** $x - \bar{x}$.
3. **Square** each deviation. The sign stops mattering, so you can use the distance from the mean.
4. **Add** the squared deviations.
5. **Divide** by $n - 1$.
6. Take the **square root**.

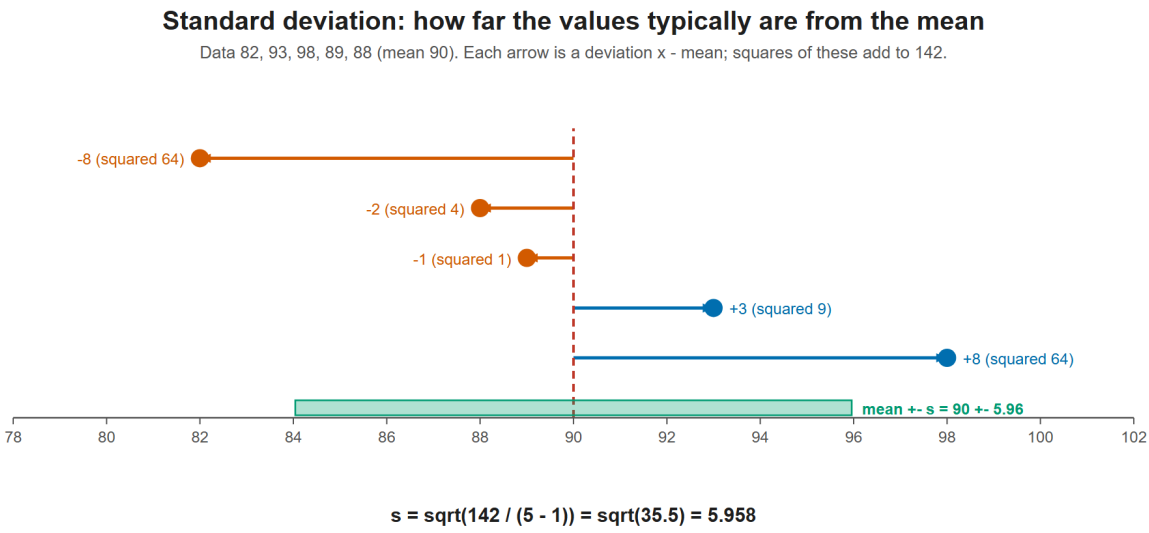
Example 1 - five values

Data: 82, 93, 98, 89, 88

Mean: $\bar{x} = \frac{450}{5} = 90$

x	$x - \bar{x}$	$(x - \bar{x})^2$
82	−8	64
93	3	9
98	8	64
89	−1	1
88	−2	4
Sum	0	**142**

$$s = \sqrt{\frac{142}{5 - 1}} = \sqrt{35.5} \approx 5.958$$



Each arrow is one deviation from the mean of 90. Squaring them (64, 9, 64, 1, 4) and adding gives 142. Dividing by n - 1 = 4 and taking the square root gives s = 5.96. The green band shows mean +- s.

Example 2 - ten values

Data: 76, 84, 69, 92, 58, 89, 73, 97, 85, 77

Mean: $\bar{x} = \dfrac{800}{10} = 80$

Shortcut: squaring removes the sign, so just write how far each value is from 80.

x	Distance from 80	Squared
76	4	16
84	4	16
69	11	121
92	12	144
58	22	484
89	9	81
73	7	49
97	17	289
85	5	25
77	3	9
Sum		**1234**

$$s = \sqrt{\frac{1234}{10-1}} = \sqrt{137.11} \approx 11.709$$

Key takeaways

- Standard deviation measures the **typical distance of the data from the mean**.
- The sample version divides by $n - 1$. The population version (σ) divides by N and uses μ.
- Check: the deviations $x - \bar{x}$ always add up to **0**.

Lesson 3.12 - Standard Deviation and Coefficient of Variation

> **Video:** Standard Deviation and Coefficient of Variation by The Organic Chemistry Tutor **Chapter:** 3 - Central Tendency

Formulas

Coefficient of variation (CV): the standard deviation as a percentage of the mean.

$$\mathrm{CV}_{\text{sample}} = \frac{s}{\bar{x}} \times 100\%, \qquad \mathrm{CV}_{\text{population}} = \frac{\sigma}{\mu} \times 100\%$$

Standard deviation:

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}, \qquad \sigma = \sqrt{\frac{\sum (x - \mu)^2}{N}}$$

A CV of 20% means the standard deviation is 20% of the mean.

Why use the CV?

- The standard deviation measures how spread out the data is around the mean.
- The CV does the same, but **relative to the mean**, and it has **no units**: in $s/\bar{x}$, feet divided by feet cancel.
- That makes the CV better for **comparing the variability of different data sets**, especially ones with different means or units.

Example - which test has more variation?

Ten students: five took test A and five took test B.

- **Test A:** 79, 84, 86, 81, 89
- **Test B:** 63, 72, 92, 84, 78

Step 1 - means

$$\bar{x}_A = \frac{419}{5} = 83.8, \qquad \bar{x}_B = \frac{389}{5} = 77.8$$

Step 2 - standard deviations

Test A:

x	$x - 83.8$	squared
79	−4.8	23.04
84	0.2	0.04
86	2.2	4.84
81	−2.8	7.84
89	5.2	27.04
Sum		**62.80**

$$s_A = \sqrt{\frac{62.8}{4}} = \sqrt{15.7} \approx 3.962$$

Test B:

x	$x - 77.8$	squared
63	−14.8	219.04
72	−5.8	33.64
92	14.2	201.64
84	6.2	38.44
78	0.2	0.04
Sum		**492.80**

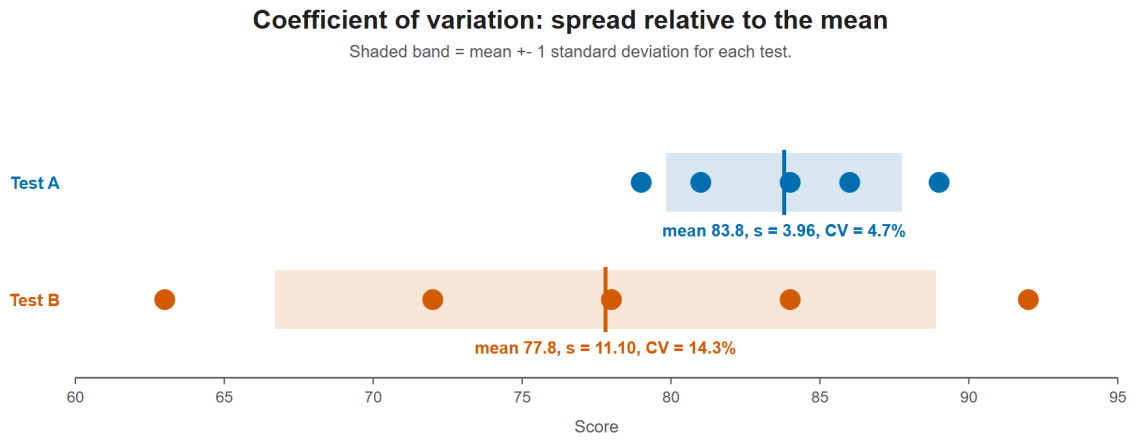
$$s_B = \sqrt{\frac{492.8}{4}} = \sqrt{123.2} \approx 11.10$$

Step 3 - coefficients of variation

$$\text{CV}_A = \frac{3.962}{83.8} \times 100\% \approx 4.7\%$$

$$\text{CV}_B = \frac{11.10}{77.8} \times 100\% \approx 14.3\%$$

Conclusion: test B's scores vary much more relative to their mean (14.3% vs 4.7%).



Test B's scores are far more spread out. Its band (mean +- s) is almost three times as wide, and relative to its mean its spread is 14.3% versus 4.7% for test A.

Key takeaways

- CV = SD ÷ mean × 100%.
- The CV is **unitless**, so it is the right tool for comparing the spread of data sets with different scales.
- Calculate the standard deviation first, then divide by the mean.

Lesson 3.13 - How to Calculate Variance

> **Video:** How To Calculate Variance by The Organic Chemistry Tutor **Chapter:** 3 - Central Tendency

Formula (sample variance)

$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$$

The variance is the **square of the standard deviation**, so it is the same calculation without the final square root.

Example 1

Data: 6, 9, 14, 10, 5, 8, 11 → sorted: 5, 6, 8, 9, 10, 11, 14 ($n = 7$)

Mean: $\bar{x} = \frac{63}{7} = 9$

x	$x - \bar{x}$	$(x - \bar{x})^2$
5	−4	16
6	−3	9
8	−1	1
9	0	0
10	1	1
11	2	4
14	5	25
Sum	0	**56**

$$s^2 = \frac{56}{7 - 1} = \frac{56}{6} \approx 9.33$$

What variance tells you

Variance measures **spread**: how far the data lies from the mean. A **larger variance means the data is more spread out**.

Example 2 - same mean, different spread

Data set 1	Data set 2
6, 7, 8, 9, 10	4, 6, 8, 10, 12
mean = 40/5 = **8**	mean = 40/5 = **8**

On a number line, both sets are centered at 8. Set 1 runs from 6 to 10, while set 2 is more spread out, from 4 to 12.

```
Set 1:          •  •  •  •  •
Set 2:    •     •     •     •     •
       ---+--+--+--+--+--+--+--+--+---
          4  5  6  7  8  9 10 11 12
```

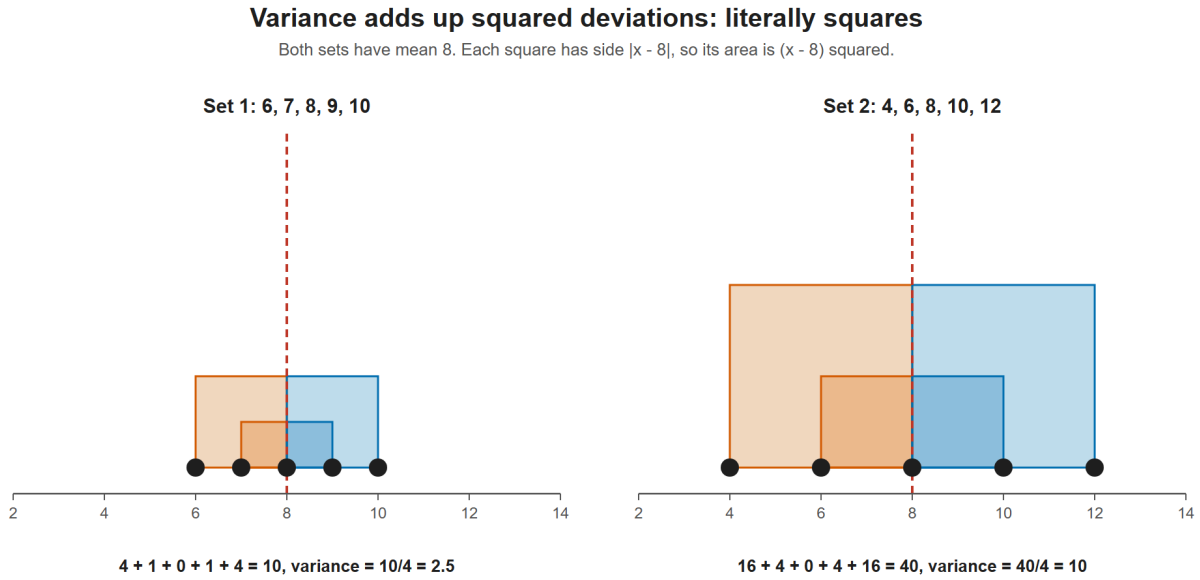
Set 1:

$$s^2 = \frac{2^2 + 1^2 + 0^2 + 1^2 + 2^2}{4} = \frac{10}{4} = 2.5$$

Set 2:

$$s^2 = \frac{4^2 + 2^2 + 0^2 + 2^2 + 4^2}{4} = \frac{40}{4} = 10$$

Set 2's variance is four times larger, because its points sit farther from the mean.



Each squared deviation is drawn as an actual square with side |x - 8|. Doubling the distances in Set 2 makes every square four times bigger, so the variance rises from 2.5 to 10.

Key takeaways

- Sample variance: square each deviation, add them, divide by $n - 1$.
- Variance = SD^2. Its units are the **square** of the data's units.
- Two data sets can have the **same mean** but **very different variances**.

Lesson 3.14 - Mean, Median, and Mode of Grouped Data

> **Video:** Mean, Median, and Mode of Grouped Data & Frequency Distribution Tables by The Organic Chemistry Tutor **Chapter:** 3 - Central Tendency

Idea

With **grouped data** you only know how many values fall in each **class interval**, not the individual values. The mean can only be **estimated**. Each class is represented by its **midpoint**:

$$\text{midpoint } m = \frac{\text{lower limit} + \text{upper limit}}{2}$$

$$\bar{x} \approx \frac{\sum f\, m}{\sum f}$$

- **Modal class:** the interval with the highest frequency.
- **Median class:** the interval containing the middle value, at position $n/2$. Find it with a **cumulative frequency** column.

Example 1 - test grades

Grade	f	Midpoint m	$f \cdot m$	Cumulative f
40-49	3	44.5	133.5	3
50-59	5	54.5	272.5	8
60-69 (D)	6	64.5	387.0	14
70-79 (C)	9	74.5	670.5	23
80-89 (B)	8	84.5	676.0	31
90-100 (A)	7	95.0	665.0	38
Total	**38**		**2804.5**	

- Midpoint example: $(40 + 49)/2 = 44.5$. The last class is $(90 + 100)/2 = 95$.

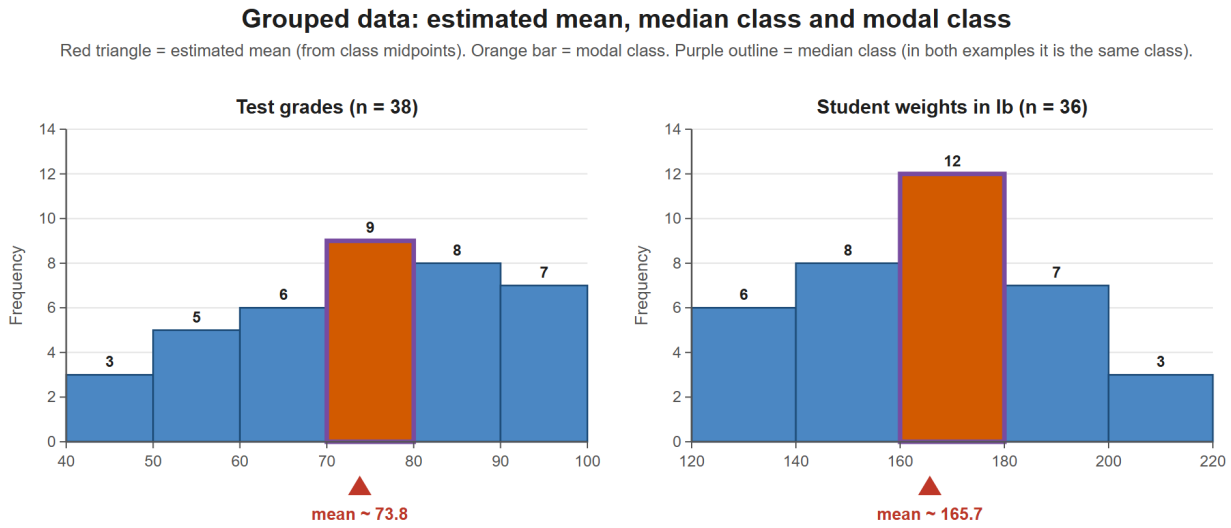
Mean:

$$\bar{x} \approx \frac{2804.5}{38} \approx 73.8$$

This is reasonable: the most common class is 70-79, and the estimate lands there.

Mode: the highest frequency is 9, so the modal class is **70-79**.

Median: half of 38 is 19. The cumulative counts reach 14 by the end of 60-69 and 23 by the end of 70-79. The 70-79 class holds students 15-23, which includes 19, so the median class is **70-79**.



The estimated mean (red triangle) is computed from class midpoints. In both examples the class with the highest bar (the modal class) also contains the middle value (the median class).

Example 2 - student weights (lb)

Weight	f	Midpoint m	$f \cdot m$	Cumulative f
120-139	6	129.5	777.0	6
140-159	8	149.5	1196.0	14
160-179	12	169.5	2034.0	26
180-199	7	189.5	1326.5	33
200-220	3	210.0	630.0	36
Total	**36**		**5963.5**	

Mean:

$$\bar{x} \approx \frac{5963.5}{36} \approx 165.7 \text{ lb}$$

The mean is in the 160-179 class but nearer 160, because more of the data lies on the lower (left) side.

Mode: the highest frequency is 12, so the modal class is **160-179**.

Median: half of 36 is 18. The classes cover students 1-6, 7-14, then **15-26**, so the 18th student is in **160-179**.

Key takeaways

- Grouped data → **estimated** mean: $\sum fm / \sum f$, using class midpoints.
- **Modal class** = highest frequency.
- **Median class** = the class whose cumulative frequency first reaches $n/2$.
- To find the exact median or mode *within* a class, see Lessons 3.16 and 3.17.

Lesson 3.15 - Standard Deviation of Grouped Data

> **Video:** How To Calculate The Standard Deviation of Grouped Data by The Organic Chemistry Tutor
> **Chapter:** 3 - Central Tendency

Formulas

For grouped data, each class is represented by its **midpoint** m and counted f times:

$$\bar{x} \approx \frac{\sum f\,m}{\sum f}, \qquad s \approx \sqrt{\frac{\sum f\,(m-\bar{x})^2}{n-1}}, \qquad n = \sum f$$

This is the ordinary sample SD formula, with each midpoint repeated f times.

Steps

1. Midpoint of each class: $m = (\text{lower} + \text{upper})/2$.
2. Multiply $f \cdot m$, then compute $\bar{x} = \sum fm / \sum f$.
3. For each class: the deviation $m - \bar{x}$, its square, then $f \cdot (m-\bar{x})^2$.
4. Add the last column, divide by $n-1$, and take the square root.

Example - exam scores (whole numbers)

The scores are whole numbers, so a value like 69.5 can't occur and the classes don't overlap.

Score	f	m	$f\,m$	$m-\bar{x}$	$(m-\bar{x})^2$	$f\,(m-\bar{x})^2$
50-59 (F)	3	54.5	163.5	−24.7	610.09	1830.27
60-69 (D)	5	64.5	322.5	−14.7	216.09	1080.45
70-79 (C)	9	74.5	670.5	−4.7	22.09	198.81
80-89 (B)	12	84.5	1014.0	5.3	28.09	337.08
90-100 (A)	8	95.0	760.0	15.8	249.64	1997.12
Total	**37**		**2930.5**			**5443.73**

Mean:

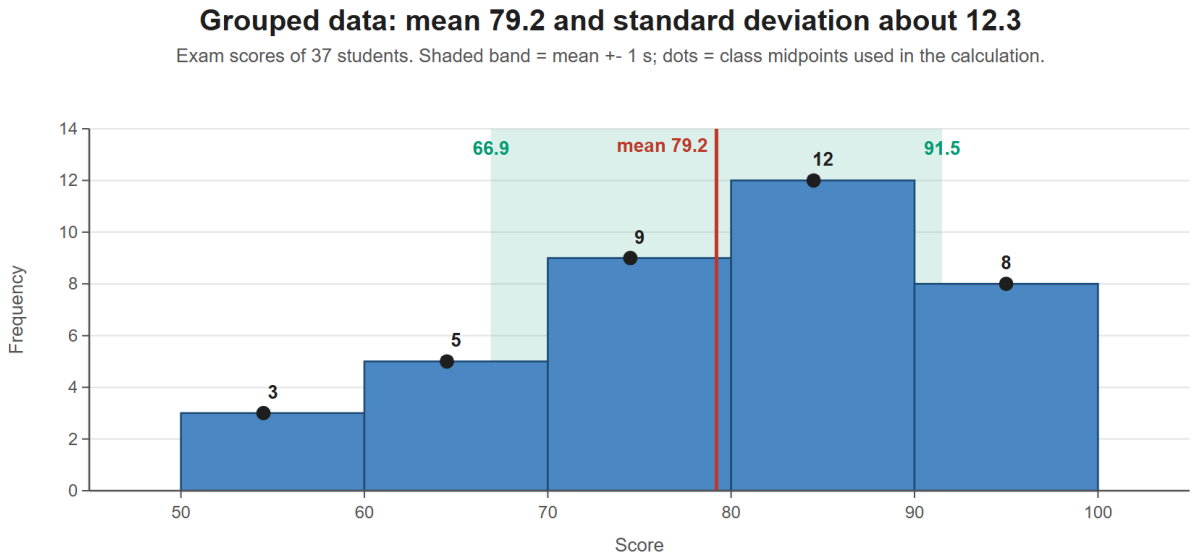
$$\bar{x} = \frac{2930.5}{37} \approx 79.2$$

This makes sense, since most students scored between 70 and 89.

Standard deviation:

$$s = \sqrt{\frac{5443.73}{37-1}} = \sqrt{151.21} \approx 12.3$$

(The deviations above use the rounded mean 79.2. Using the unrounded mean changes the answer only in later decimals.)



Each class is represented by its midpoint (black dots) when computing the mean and standard deviation. The shaded band from 66.9 to 91.5 is the mean +- 1 standard deviation.

Key takeaways

- Replace each class by its **midpoint**, weighted by its **frequency**.
- $s \approx \sqrt{\sum f(m - \bar{x})^2 / (n - 1)}$ with $n = \sum f$.
- The result is an **estimate**, because the actual values inside each class are unknown.
- A spreadsheet or calculator makes the columns much faster to fill in.

Lesson 3.16 - How to Calculate the Median of Grouped Data

> **Video:** How To Calculate the Median of Grouped Data by The Organic Chemistry Tutor **Chapter:** 3 - Central Tendency

Formula

$$\text{median} = L + \left(\frac{\frac{n}{2} - C}{f}\right) h$$

Symbol	Meaning
L	**Lower boundary** of the median class
n	Total frequency, $\sum f$
C	**Cumulative frequency of the class before** the median class
f	Frequency of the median class
h	Class width, **upper boundary − lower boundary**

Class boundaries vs class limits

If the classes are written 60-69, 70-79, ..., there is a gap between 69 and 70. The **boundaries** sit halfway across each gap:

- lower boundary of 70-79 = $\dfrac{69 + 70}{2} = 69.5$
- upper boundary of 70-79 = $\dfrac{79 + 80}{2} = 79.5$
- width $h = 79.5 - 69.5 = 10$, **not** $79 - 70 = 9$

Example

Grade	f	Cumulative f
40-49	3	3
50-59	5	8
60-69	6	**14** ← C
70-79	**9**	23 ← median class
80-89	8	31
90-100	7	38

Step 1 - find the median class. $n = 38$ and $n/2 = 19$. The classes hold the scores in positions 1-3, 4-8, 9-14 and **15-23**, ... The 19th value is in the **70-79** class.

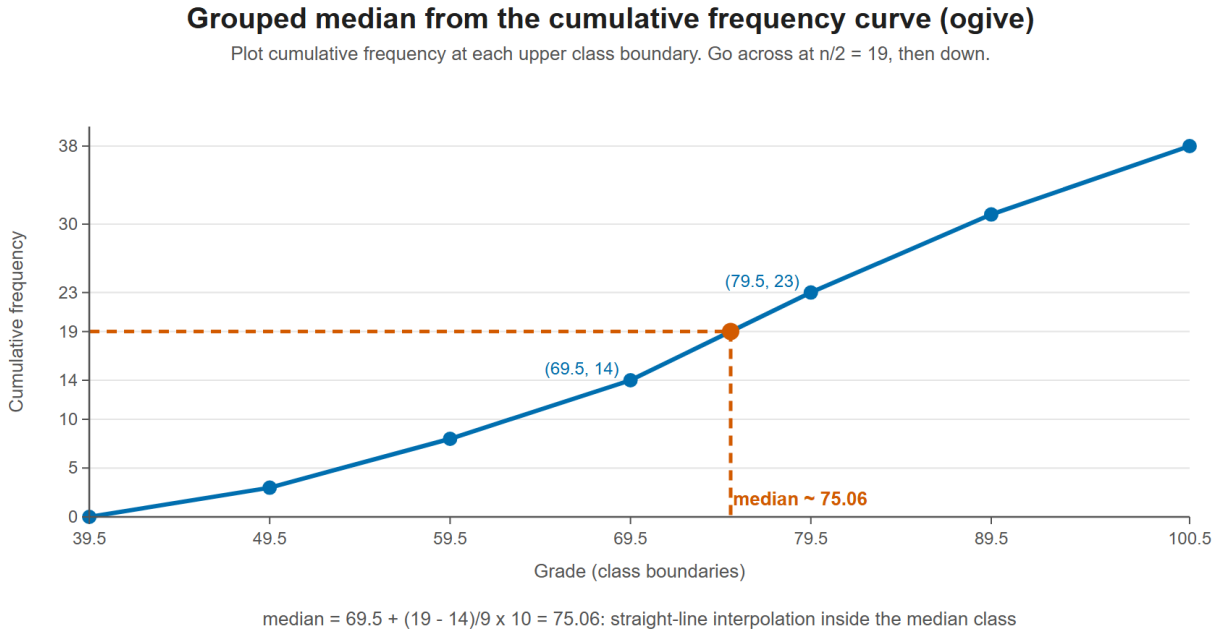
Step 2 - collect the values. $L = 69.5$, $C = 14$, $f = 9$, $h = 10$.

Step 3 - substitute.

$$\text{median} = 69.5 + \left(\frac{19 - 14}{9}\right)(10) = 69.5 + \frac{50}{9} \approx 69.5 + 5.56 = 75.06$$

The median is about **75.1**, which is inside the median class 70-79 as it should be.

> **Correction to the video:** the video gives 75.5 (repeating) as the result. But $50/9 \approx 5.56$ and $69.5 + 5.56 \approx 75.06$, so the correct median is about **75.06**.



The formula is the same as reading the cumulative frequency curve. Go across from n/2 = 19 to the curve, then straight down. The median lands about 5/9 of the way through the 69.5-79.5 class.

Key takeaways

- Find the **median class** first, using the cumulative frequency to locate position $n/2$.
- Use class **boundaries** (69.5, 79.5) for L and h when there are gaps between class limits.
- C is the cumulative frequency **before** the median class, not including it.
- Check that the answer lies inside the median class.

Lesson 3.17 - How to Calculate the Mode of Grouped Data

> **Video:** How To Calculate the Mode of Grouped Data by The Organic Chemistry Tutor **Chapter:** 3 - Central Tendency

Formula

$$\text{mode} = L + h\left(\frac{f_m - f_1}{2f_m - f_1 - f_2}\right)$$

Symbol	Meaning
L	**Lower boundary** of the modal class
h	Class width (upper boundary − lower boundary)
f_m	Frequency of the **modal class** (the highest frequency)
f_1	Frequency of the class **before** the modal class
f_2	Frequency of the class **after** the modal class

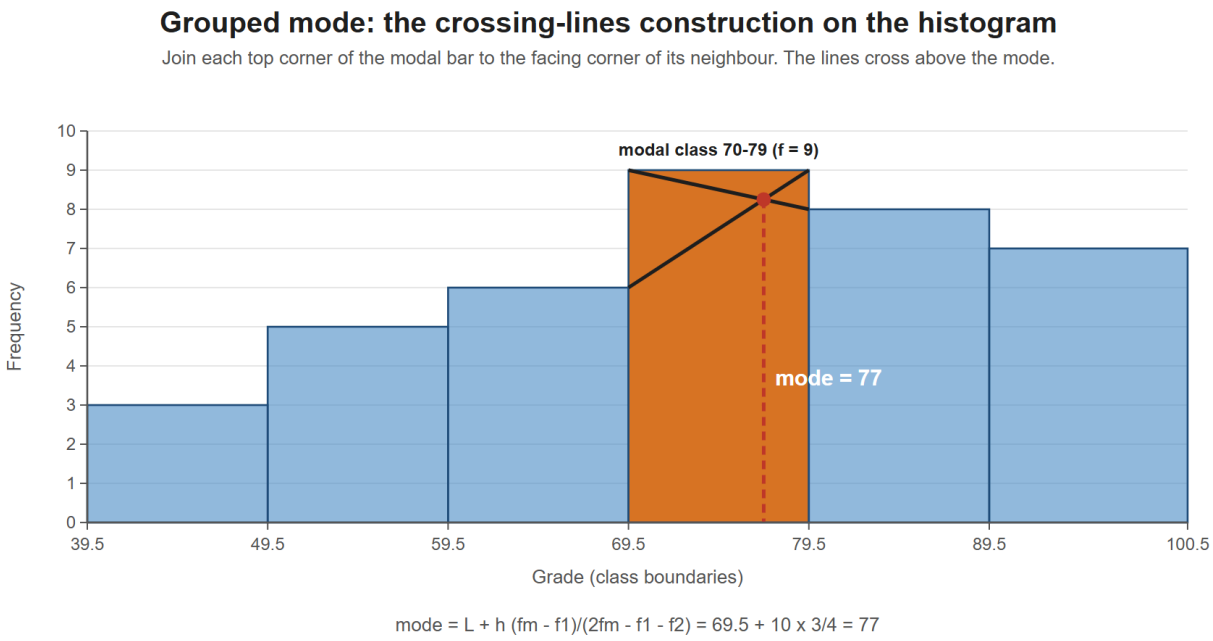
Example

Grade	f
40-49	3
50-59	5
60-69	6 ← f_1
70-79	**9** ← f_m (modal class)
80-89	8 ← f_2
90-100	7

Values. $L = 69.5$ (halfway between 69 and 70), $h = 79.5 - 69.5 = 10$, $f_m = 9$, $f_1 = 6$, $f_2 = 8$.

$$\text{mode} = 69.5 + 10\left(\frac{9-6}{2(9)-6-8}\right) = 69.5 + 10\left(\frac{3}{4}\right) = 69.5 + 7.5 = 77$$

Does it make sense? 77 lies in the modal class 70-79, and it is nearer 79 than 70. The class after the modal class has a higher frequency (8) than the class before it (6), so the mode is pulled toward the upper end.



A graphical version of the formula. The two lines joining the modal bar to its neighbours cross at grade 77. The mode leans toward 79.5 because the next class (8) is taller than the previous one (6).

Key takeaways

- The modal class is the one with the **highest frequency**.
- Use **boundaries** for L and h: 69.5 and a width of 10, not 70 and 9.
- The formula shifts the mode toward whichever neighboring class has the larger frequency.

Lesson 3.18 - Mean Absolute Deviation

> **Video:** Mean Absolute Deviation by The Organic Chemistry Tutor **Chapter:** 3 - Central Tendency

Idea

The **mean absolute deviation (MAD)** is the **average distance** of the data values from the mean.

$$\text{MAD} = \frac{\sum |x - \bar{x}|}{n}$$

Steps

1. Find the mean $\bar{x}$.
2. Find each deviation $x - \bar{x}$.
3. Take **absolute values**, so all distances are positive.
4. Add them and divide by n.

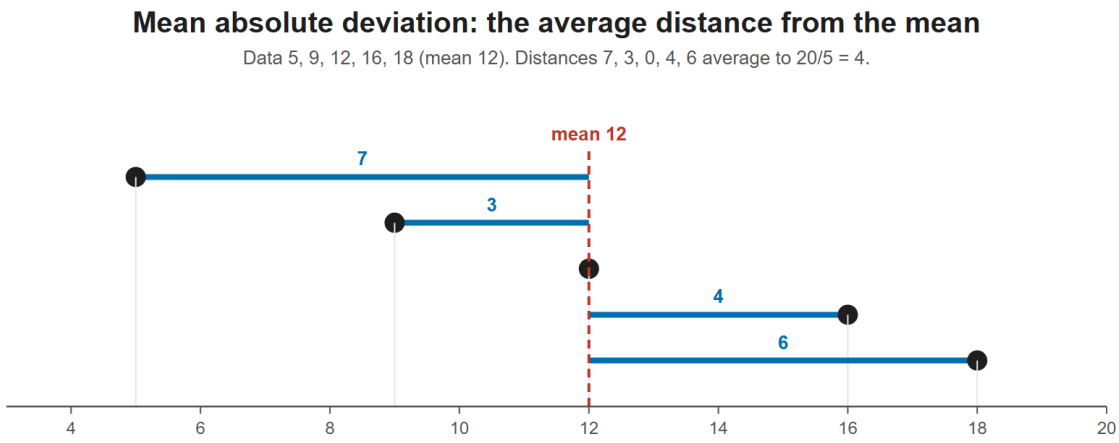
Example 1 - with a table

Data: 7, 11, 14, 19, 22, 29

Mean: $\bar{x} = \frac{102}{6} = 17$

x	$x - \bar{x}$	$\lvert x - \bar{x} \rvert$
7	−10	10
11	−6	6
14	−3	3
19	2	2
22	5	5
29	12	12
Sum	0	**38**

$$\text{MAD} = \frac{38}{6} = 6\tfrac{1}{3} \approx 6.33$$



Each bar is the distance of one value from the mean of 12. The mean absolute deviation is the average bar length, (7 + 3 + 0 + 4 + 6)/5 = 4.

Example 2 - directly with the formula

Data: 5, 9, 12, 16, 18

Mean: $\bar{x} = \dfrac{60}{5} = 12$

$$\text{MAD} = \frac{|5-12| + |9-12| + |12-12| + |16-12| + |18-12|}{5} = \frac{7+3+0+4+6}{5} = \frac{20}{5} = 4$$

What it means

Each absolute deviation is a distance from the mean 12 on the number line:

Value	5	9	12	16	18
Distance from 12	7	3	0	4	6

The MAD is the average of these five distances: $(7+3+0+4+6)/5 = 4$. On average, each value is **4 units** from the mean.

Key takeaways

- MAD = average of the **absolute** deviations from the mean.
- Absolute values are needed because the plain deviations always add up to **zero** (see Lesson 3.19).
- MAD is easier to interpret than the standard deviation (no squaring), but the SD is used more often in later statistics.

Lesson 3.19 - How to Calculate the Average Deviation

> **Video:** How To Calculate the Average Deviation by The Organic Chemistry Tutor **Chapter:** 3 - Central Tendency

Formula

The **average deviation** (signed, with **no** absolute value) is

$$A = \frac{\sum(x - \bar{x})}{n}$$

Compare it with the **mean absolute deviation** (Lesson 3.18), which uses $|x - \bar{x}|$.

Example

Data: 8, 12, 15, 18, 19, 23, 31

Mean: $\bar{x} = \frac{126}{7} = 18$

Deviations:

x	8	12	15	18	19	23	31
$x - 18$	−10	−6	−3	0	1	5	13

$$A = \frac{-10 - 6 - 3 + 0 + 1 + 5 + 13}{7} = \frac{0}{7} = 0$$

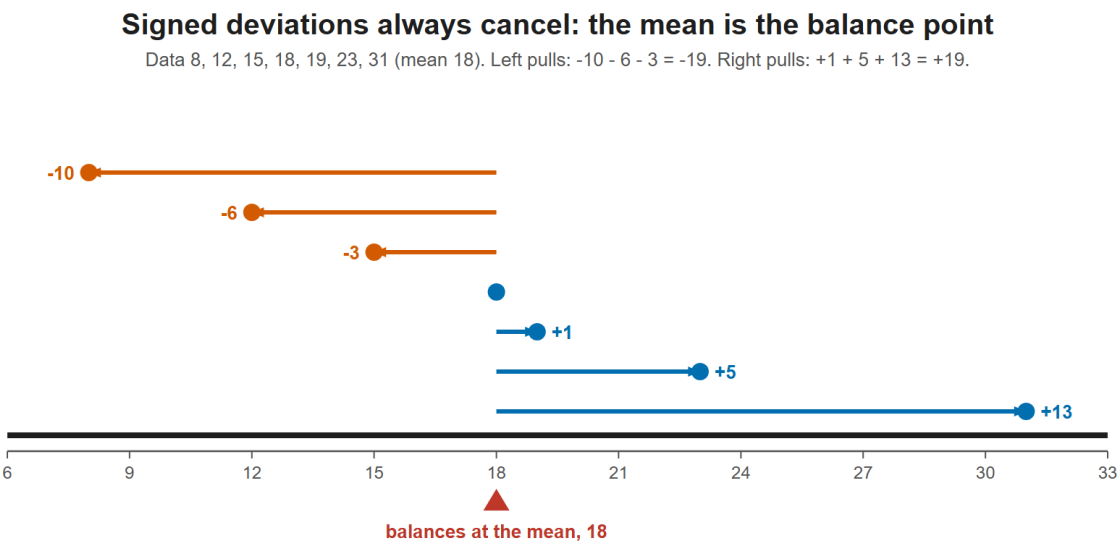
The positives and negatives cancel: -6 with $1 + 5$, and $-10 - 3 = -13$ with $+13$.

Why is it always zero?

$$\sum(x - \bar{x}) = \sum x - n\bar{x} = \sum x - \sum x = 0$$

This holds for **every** data set: the mean is exactly the balance point of the data. So the signed average deviation **cannot** measure spread, which is why statistics uses:

- the **mean absolute deviation**, which uses absolute values, and
- the **variance / standard deviation**, which square the deviations.



The leftward pulls (-10, -6, -3) add to -19 and the rightward pulls (+1, +5, +13) add to +19, so the signed deviations always total zero. The mean is the point where the data balance.

Key takeaways

- Average (signed) deviation = $\sum(x - \bar{x})/n$, and it is **always 0**.
- It is a useful check: if your deviations don't add to 0, the mean is wrong.
- To measure spread, use the MAD or the standard deviation instead.